

- Q.5 HMI in SCADA stands for :
- a) Hardware Machine Interface
 - b) Human Monitoring Interface
 - c) Human Machine Interface
 - d) High Monitoring Indicator
- Q.6 In SCADA, RTU stands for :
- a) Real Time Unit
 - b) Remote Terminal Unit
 - c) Reactive Terminal Unit
 - d) Remote Transmission Unit
- Q.7 Profibus is mainly used for :
- a) Wireless communication
 - b) Point to point analog communication
 - c) High speed field level communication
 - d) Audio transmission
- Q.8 Relay works on the principle of
- a) Ohm law
 - b) Hooks law
 - c) Electrically
 - d) Electromagnetic Induction
- Q.9 Which of the following best describes the architecture of SCADA?
- a) Decentralized control only
 - b) Centralized data collection with remote field devices
 - c) One-to-one connection
 - d) Peer to peer without supervision

- Q.10 SCADA is more suitable for :
- a) Large scale geographical monitoring systems
 - b) Local machine control
 - c) Manual data entry
 - d) Internal PLC logic

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Expand SCADA.
- Q.12 Define Field bus.
- Q.13 Explain Scan cycle.
- Q.14 Explain HMI.
- Q.15 What is the difference between Switches and relays.
- Q.16 What is the purpose of buses used in Instrumentation.
- Q.17 Explain move instruction used in plc programming.
- Q.18 Expand HART.
- Q.19 Ladder Logic is a common language used to program PLCs (True/False)
- Q.20 Explain industrial automation.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the all elements of SCADA system.
- Q.22 What are the difference between DCS and SCADA system?
- Q.23 Explain the all buses used in Instrumentation system.
- Q.24 What are the salient features of Ethernet?